Ruby testing has historically been dominated by RSpec, but Minitest has made a resurgence in recent years. Minitest is rooted in providing a minimal set of assertions so that tests remain simple. RSpec rooted itself in a more comprehensive assertion library and has many options for setting up data.

Elixir's testing framework—ExUnit—is included with the language. You don't need to install anything to use it. In some languages, the built-in testing framework isn't great, but ExUnit is actually good. (This is evidenced by the fact that there isn't another mainstream testing framework in Elixir.)

ExUnit is similar to Minitest in that it uses a few simple assertions and has minimal options for setting up test cases. This minimal approach has proven to be powerful for ExUnit. It's quick to learn, and you'll often find that you have everything you need to write clean, expressive tests.

Let's dive in!

Test NewMessage Changeset

The first test we're going to write is for an Ecto changeset. This is a great first test to write because the test doesn't involve database or web requests. All that we need to do is call a function and check the output of it.

Let's verify that your environment is correctly set up to run tests. Type `mix test` and verify that your existing tests are green:

```
$ mix test
.....
Finished in 0.08 seconds (0.04s async, 0.04s sync)
5 tests, 0 failures
```

These five tests are the default tests included by Phoenix. You could remove them, but we'll keep them for now and move on.

Your tests should be green without any extra steps. If you do run into an error, it would likely be around the database setup. If this occurs, run `MIX_ENV=test mix do ecto.drop, ecto.create, ecto.migrate` to reset your test database. (You can make an alias for this in `mix.exs`, but it isn't included by default.)

Now let's test the changeset. Create the file `test/phone_app/conversations/schema/new_message_test.exs` and add the following code:

phone_app/test/phone_app/conversations/schema/new_message_test.exs

```
defmodule PhoneApp.Conversations.Schema.NewMessageTest do
  use ExUnit.Case, async: true

  alias PhoneApp.Conversations.Schema.NewMessage
end
```